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AI-Based Parcel Detection System

AI-Solution Documentation

Relevance of the AI Solution to the Theme

The proposed AI-Based Parcel Damage Detection System is relevant to the theme "An AI Solution for industries" because it applies artificial intelligence to improve an operational process within the e-commerce industry. The system uses computer vision and machine learning to inspect the external condition of parcels inside the Takealot warehouse before they are dispatched. By automatically identifying damaged and intact parcels, the proposed solution can assist warehouse staff in identifying parcels that require inspection before dispatch. This demonstrates how AI can be applied in an industry environment to support business operations, improve quality control, and create a record of parcel conditions before dispatch. As part of the fourth industrial revolution, this applies AI to the e-commerce and logistics industry, where fast and accurate quality control directly affects customers.

2. Business Objectives and Requirements

2.1 Business Background

Takealot is an e-commerce company that processes and distributes many customer orders. The packaging and dispatch process involves preparing parcels before they are sent to customers. During this process, parcels may have visible external packaging damage such as dents, tears, holes, or crushed areas. If a parcel is dispatched with existing damage, it may later be reported as damaged by the customer, making it difficult to determine when the damage occurred. The proposed AI solution aims to introduce an automated inspection point inside the Takealot warehouse. This can provide a record of the visible condition of the parcel at the point of dispatch.

2.2 Business Objectives

- The main objective is to automatically classify parcels as Damaged or Intact before dispatch, so that damaged parcels are identified and recorded before they leave the warehouse.

- Automatically inspect the visible external condition of parcels before dispatch.

  Identify parcels that show visible signs of packaging damage.

- Classify parcels as **Damaged** or **Intact** using an AI-based image classification system.
- Assist warehouse staff in identifying damaged parcels before they leave the warehouse.
- Reduce the possibility of visibly damaged parcels being dispatched without inspection.
- Create a visual record of the parcel's condition before dispatch that can support future investigations of damage related complaints.
- Improve the efficiency of the parcel inspection process by assisting staff with automated image analysis.

2.3 Business Success Criteria

The proposed system would be considered successful if it can:

- Correctly classify parcel images as damaged or intact with an accepted level of at least 90% accuracy.
- Detect common forms of visible external packaging damage including dents, tears, holes and crushed areas.
- Provide classification results quickly enough to support the parcel inspection process.
- Reduce the need for staff to manually inspect every parcel for visible packaging damages.
- Produce a record of the parcel's condition before dispatch.

The final level of accuracy and performance would need to be determined through testing using suitable evaluation data.

2.4 Business Requirements

- The system must capture or receive an image of a parcel's external packaging.
- The system must analyse the parcel image using an AI-based image classification model.
- The system must classify the parcel as either **Damaged or Intact.**

  The system should provide a confidence value for its classification were supported by the model.
- Parcels classified as damaged should be flagged for further inspection by warehouse staff.
- Parcels classified as intact should be allowed to continue through the normal dispatch process.
- The system should maintain or support the creation of a visual record of the parcel's condition before dispatch.
- The system should be able to process images representing different types of parcel packaging and visible damage conditions.

2.5 Constraints

- The system depends on the quality and availability of parcel images.
- Poor lighting, camera positioning, image quality, or objects blocking the packaging may affect classification accuracy.
- The training dataset may not contain every possible type of packaging or damage found in a real warehouse.
- The system is designed to identify visible external packaging damage and cannot determine whether a product inside a parcel is damaged.
- The proposed system is focused on the inspection of parcels inside the warehouse before dispatch and does not determine where damage occurred during the delivery process.
- The performance of the AI model depends on the quality, quantity and variety of training data available.

2.6 Risks

- **False positive:** An intact parcel may be incorrectly classified as damaged, causing unnecessary inspection.
- **False negative:** A damaged parcel may be incorrectly classified as intact and continue with the dispatch process.
- **Poor image quality**: Blurry, dark or poorly positioned images may result in incorrect classifications.

  Limited training data: A dataset that does not represent enough packaging types and damage conditions may reduce the model's ability to generalise to new images.
- **Environmental changes**: Changes in lighting, camera position or the appearance of parcels may affect model performance.
- **Reliance on AI**: Staff should still be able to inspect flagged parcels because AI predictions may not always be correct.

2.7 Initial Assessment of Tools and Techniques

The proposed solution will use Python as the main programming language because it provides libraries and frameworks suitable for developing AI and image-processing applications, such as TensorFlow/Kreas, NumPy, Pillow and Matplotlib. Computer vision techniques will be used to analyse images of parcel packaging. Machine learning and deep learning techniques, particularly image classification using a Convolutional Neural Network {CNN} will be used to distinguish between damaged and intact parcels. Transfer learning with the pre-trained MobileNetV2 model will be used because it is lightweight, accurate and performs well on a small dataset. A labelled dataset containing images of damaged and intact packaging will be used to train and evaluate the model. Image preprocessing such as resizing and scaling the images and data augmentation such as shear, zoom and flipping will be applied to improve the model's ability to recognise different packaging conditions. The model's performance will be evaluated using accuracy, precision, recall, F1-score and a confusion matrix.

3. Problem Definition

Takealot processes a large volume of customer orders that are packaged and dispatched from its warehouse. During the packaging and dispatch process, parcels may have visible external damage such as dents, holes, tears and crushed areas. When this damage is not identified and recorded before dispatch, a parcel may leave the warehouse in a damaged condition, and the issue may only become known when the customer reports it after delivery. This makes it difficult for Takealot to determine whether the damage occurred before the parcel was dispatched or during transportation. It can also result in additional business costs associated with refunds, replacements, returns and investigations, while potentially affecting customer satisfaction and trust. The absence of an automated inspection process makes it difficult to consistently identify and record the condition of every parcel before it leaves the warehouse.

This problem is relevant to the theme because AI-based computer vision can automate quality control in the e-commerce industry. The proposed AI solution uses image classification to automatically analyse parcel images before dispatch and classify them as either damaged or intact. This benefits Takealot, as it allows staff to inspect and repackage parcels earlier and to create evidence of parcel condition before dispatch. It also benefits customers, who are more likely to receive parcels in good condition. Using AI therefore supports quality control, reduces avoidable damage-related costs and improves the efficiency of the warehouse dispatch process.

AI Solution Theoretical Aspect

1.Machine Learning Approach

The system uses supervised machine learning for binary image classification. It is supervised because every training image carries a label, either damaged or intact, and the model learns the visual patterns that separate the two classes. The solution was developed in two stages. First, a small Convolutional Neural Network (CNN) was trained from scratch as a baseline. It scored only about 52% on the test set, because the dataset is small. The final model therefore uses transfer learning with MobileNetV2, which reuses visual features learned from a very large image dataset and adapts them to parcel packaging. This raised the best validation accuracy to approximately 91%. MobileNetV2 was chosen because it is lightweight, accurate and well suited to small datasets and to devices in a warehouse. The overall process is collecting labelled images, preprocess them, train, validate, test and deploy the model in a web application. The model learns visual patterns from labelled images and predicts the category of a new image, helping warehouse staff to identify parcels that require further inspection.

2.Data

The model uses labelled images of damaged and intact parcel packaging. The dataset contains 349 training images, 100 validation images and 27 test images, arranged in folders by class (damaged and intact) within each of the three sets. Images are converted to RGB, resized to 160×160 pixels and scaled before classification. During training, data augmentation (shear, zoom and horizontal flipping) creates extra variation. The forms of data relevant to this solution are:

- **Training data:** labelled images of damaged and intact packaging, used to teach the model.
- **Validation and test data:** separate labelled images used to monitor and measure performance on unseen examples.
- **Live input data:** photos of parcels uploaded or captured with a camera when the application is used.
- **Output data:** the predicted class (damaged or intact) and a confidence value.

- **Inspection records (planned):** the date, prediction and staff verification for each parcel, which would support the time series analysis.

3.Model Evaluation

The final MobileNetV2 model is evaluated on two separate sets. The validation data is used during training to monitor accuracy and loss and to detect overfitting. The test data is kept aside and used only once at the end, to measure how the model performs on images it has never seen. The model achieved approximately 91% best validation accuracy and 100% accuracy on the test set (27 out of 27 images). This compares with about 52% for the baseline CNN, which shows the benefit of transfer learning, and it meets the success criterion of at least 90% accuracy. Because the test set is small, this result will be confirmed on a larger set of images. Accuracy alone does not show what kind of mistakes the model makes, so the model will also be assessed with precision, recall, the F1-score and a confusion matrix. Recall is particularly important, because it measures how many damaged parcels the model correctly identifies, and a damaged parcel classified as intact would continue through dispatch.

4.Time Series Analysis on Data

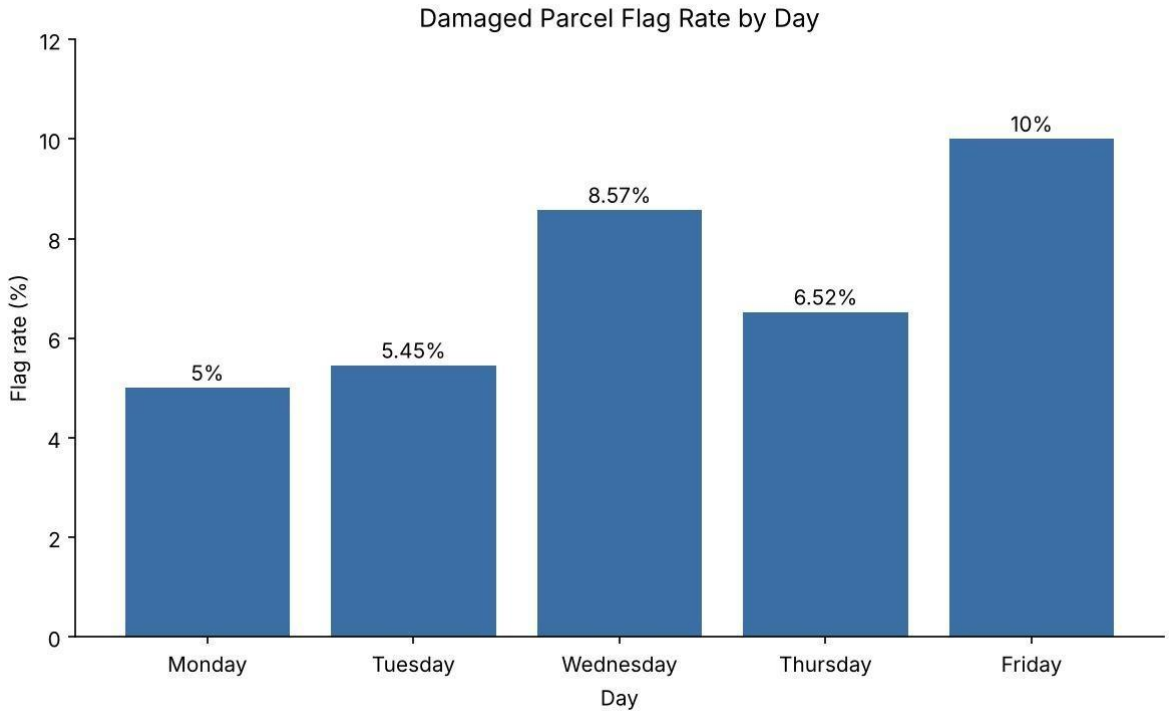
Time series analysis examines data collected over time to identify trends and changes. For this project, inspection records could be grouped by day to monitor the number and percentage of parcels flagged as damaged. The current application classifies individual images so this analysis would require an additional feature to store inspection dates, predictions and verification outcomes.

The damaged flag rate would be calculated as follows:

Damage-flag rate (%) = (Parcels flagged as damaged) divided by (Total parcels inspected) *100

The following figures are hypothetical and illustrates how the analysis could work

Day	Parcel Inspection	Flagged	Flag rate
Monday	200	10	5%
Tuesday	220	12	5.45%
Wednesday	210	18	8.57%
Thursday	230	15	6.52%
Friday	250	25	10%



A bar graph could display these daily rates and highlight increases or decreases. For example, the rise from 5% on Monday to 10% on Friday could prompt warehouse management to investigate possible packaging or handling issues. This analysis could help monitor **packaging** quality and support better informed warehouse decisions.

5.Solution Techniques

Several techniques are used to find a solution and to improve the accuracy of the model. Image resizing and data augmentation prepare the images and introduce variation during training. MobileNetV2 transfer learning reuses pre-trained visual features, which gave a large improvement over the baseline model. Dropout and early stopping reduce overfitting, and evaluating the model on separate validation and test sets helps to identify weaknesses and to assess its ability to classify unseen parcel images. Increasing the image size from 64×64 to 160×160 pixels also preserved the detail of small dents and tears.

The following techniques are planned to improve accuracy further:

- Fine-tuning the top layers of MobileNetV2 with a low learning rate.
- Collecting more and varied images, especially of difficult cases such as poor lighting and different camera angles.
- Reviewing misclassified images and retraining the model on similar examples.
- Tuning hyperparameters such as the learning rate, dropout rate and size of the dense layer.
- Adjusting the decision threshold so that damaged parcels are less likely to be missed

6.Natural Language Processing (NLP), Speech Recognition or Speech Synthesis

A speech synthesis feature is proposed for the system. After a parcel is classified, the application would read the result aloud, for example “Damaged parcel detected, please inspect.” This is relevant to the theme because warehouse staff often work with their hands full and cannot always look at a screen. It is achievable with free Python libraries such as gTTS or pyttsx3, and it only requires a few extra lines of code added to the existing application, with no change to the trained model. Speech recognition could also be added, using a library such as Speech Recognition, so that staff can start a scan with a voice command such as “scan parcel”.

7.Deep Learning

The solution is based on deep learning. The final model uses MobileNetV2, a convolutional neural network architecture, with transfer learning. The deep learning techniques used are:

- **Convolutional layers**, which learn visual features such as edges, textures and creases directly from the images.
- **Depthwise separable convolutions, inverted residual blocks and batch normalisation**, which make MobileNetV2 accurate and efficient.
- **Transfer learning**, in which the pre-trained layers are frozen and used to extract visual features.
- **Global average pooling**, which condenses the extracted features before classification.
- **Dense layers with ReLU activation and dropout**, which form the new classification layers.
- **A sigmoid output**, which predicts the probability that a parcel is intact or damaged.
- **Binary cross-entropy loss, the Adam optimizer and backpropagation**, which train the model.
- **Data augmentation and early stopping**, which reduce overfitting.

Future improvements include fine-tuning the pre-trained layers and using an object detection model, such as YOLO, to show exactly where on the parcel the damage is. This approach is valuable because deep learning can learn the visual patterns associated with external packaging damage directly from labelled images, which allows parcel inspection to be automated.

8. Chatbot/Softbot

A chatbot is proposed as an additional feature of the system. It would be added as a separate tab in the existing Streamlit application and would receive the model's prediction as input. It could answer warehouse staff questions about the inspection process, for example "What should I do with a damaged parcel?" and explain what action to take when a parcel is flagged, such as inspecting it, repackaging it and recording the result. It could start as a rule-based chatbot with predefined questions and answers and later use a large language model for open-ended questions. It could provide consistent guidance while leaving the physical inspection and final decision to staff.